NATIONAL INSTITUTES OF HEALTH (NIH)

Your Guide to Lowering Your Blood Pressure with DASH Eating Plan

National Heart, Lung, and Blood Institute (NHLBI / NIH Publication No. 06-4082)
Official Reference: NIH Publication No. 06-4082 / NHLBI Clinical Trials
Official Web Portal: https://www.nhlbi.nih.gov/education/dash-eating-plan

1. Clinical Evidence & Efficacy of the DASH Plan

* DASH stands for Dietary Approaches to Stop Hypertension, developed through NIH multicenter clinical trials
* Lowers systolic blood pressure by 8 to 14 mmHg within 2 weeks of initiation.
* Efficacy is equivalent to a full-dose single antihypertensive drug; reduces coronary events by >25%.

2. The 8 Food Group Servings (2,000 kcal Daily Standard)

Food Group	Servings	Standard Serving Size	Health Benefit
Whole Grains	6-8 / day	1 slice whole wheat bread, 1/2 c cooked brown rice	Energy, fiber, magnesium
Vegetables	4-5 / day	1 c raw leafy greens, 1/2 c cooked vegetables	Potassium, magnesium, fiber
Fresh Fruits	4-5 / day	1 medium fruit, 1/2 c fresh/frozen fruit	Potassium, vitamin C, fiber
Low-Fat Dairy	2-3 / day	1 c skim milk, 1 c low-fat yogurt, 1.5 oz low-fat cheese	Calcium, high-quality
Lean Meat/Fish	<=6 oz / day	~170 g cooked skinless poultry, wild salmon	Lean protein, iron, zinc
Nuts & Legumes	4-5 / week	1/3 c nuts (almonds/walnuts), 1/2 c cooked beans	Magnesium, healthy fats
Healthy Fats	2-3 / day	1 tsp extra virgin olive oil, 1 tsp canola oil	Monounsaturated fats
Added Sweets	<=5 / week	1 tbsp sugar or jam, 1/2 c fruit sorbet	Sugar restriction

3. The Golden Mineral Triad in Blood Pressure Regulation

* Potassium: Promotes renal natriuresis (sodium excretion) and induces vascular smooth muscle relaxation.
* Magnesium: Modulates vascular tone, reduces peripheral resistance, and improves endothelial responsiveness
* Calcium: Regulates vascular contraction dynamics and membrane stabilization.

4. Sodium Restriction Limits

* Standard DASH Sodium Limit: Up to 2,300 mg of sodium daily (approx. 1 level teaspoon of table salt = 5 g).
* Lower Sodium DASH Target: Up to 1,500 mg of sodium daily (approx. 2/3 level teaspoon of salt).
* The 1,500 mg limit provides superior blood pressure reduction, particularly in hypertensive adults.

5. Hidden Dietary Sodium Warning (>75% of Total Daily Intake)

* Processed deli meats (sausages, salami, pastrami, frankfurters).
* Canned soups, commercial broths, and bouillon cubes.

* Pickled vegetables, brine-cured olives, and heavily salted cheeses (feta, halloumi).
* Commercial sauces: Soy sauce, ketchup, barbecue sauce, and ready-made salad dressings.